



MLOps & Model Deployment — What to Expect

A hands-on, 3-day workshop for UG students (CSE · AI & DS · IT)

Over three days you'll take **one machine-learning model all the way to a live, monitored deployment** — train it, track it, wrap it in an API, containerize it, automate it, push it to a public URL, and watch it for drift. This is a build-along workshop: your laptop is open the whole time, and by day 3 you'll have a working model on the internet you can put on your CV.

DURATION 3 continuous days · 5 hrs/day	DATES [to be confirmed]	VENUE University campus — A/C classroom + lab
AUDIENCE UG · CSE / AI & DS / IT (up to 30)	PREREQUISITES Basic Python; intro ML	COST TO YOU Free tools only

What you'll learn

Understand the ML lifecycle and why models fail after training; track experiments and version models (MLflow); package and serve a model as a REST API (FastAPI); containerize and automate it (Docker, GitHub Actions); deploy it to a free public host; and monitor it for drift with a plan to retrain.

The pipeline you'll build (one project, all 3 days)

1	Train & track	Train a model and log every run to MLflow; register & version the best one.
2	Serialize & serve	Freeze the model and expose it via a FastAPI REST API with input validation.
3	Containerize	Package the API in a Docker image that runs anywhere.
4	Automate (CI/CD)	Lint, test and build on every push with GitHub Actions — plus a model accuracy gate.
5	Deploy	Ship the container to a free host (Hugging Face Spaces / Render) — a public URL.
6	Monitor	Detect data drift, alert, and define a retraining trigger.

How it runs & how you're graded

Hands-on & interactive. ~70% of class time is live-coding and labs. You'll pair up, take on "break it" challenges, and finish on a demo day.

ASSESSMENT A short **quiz**, a **live demo** of your deployed model, and a **mock technical interview** with the industry expert.

What you need to arrange — before Day 1

PRE-CLASS CHECKLIST (PLEASE COMPLETE BEFOREHAND)

- **Bring a laptop** (8 GB RAM+, ~10 GB free) and charger. You'll install and run Docker locally.
- **Install Python 3** and **Docker Desktop**; check `docker run hello-world` works.
- **Create free accounts:** **GitHub** (github.com) and **Hugging Face** (huggingface.co) — both free, no card.
- **Basic Python & ML** is enough — you'll get starter code; you wire the pieces together.
- **Come ready to ship.** Bring curiosity and a bit of patience for the first Docker build.